Vaibhav Dasarahalli Ashoka

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SUMMARY

Python, FastAPI, LangChain and RAG engineer with 2+ years building production-grade LLM and NLP systems in financial services at OCBC Bank. Delivered measurable business outcomes: $2M+ in operational cost savings, SGD 30K/month in new revenue, and 90% reduction in manual screening workload. Specializes in multi-agent AI systems, RAG pipelines, and end-to-end ML deployment, with a track record of turning complex data problems into scalable, impactful products.

EXPERIENCE

Data Scientist | OCBC Bank Pte. Ltd | Singapore | December 2023 - June 2026

- Achieved ~90% reduction in manual screening workload by leading end-to-end development of AI-driven risk and compliance platforms, from system design through deployment and monitoring.
- Architected scalable ML pipelines and APIs (FastAPI, Docker) processing 1M+ events/week at 99.9% uptime in an Agile environment.
- Improved data accuracy to ~95% by solving complex entity extraction and document understanding problems using LangChain + Llama.
- Boosted testing and reporting efficiency by ~70% through large-scale data analysis with Python and Pandas to surface trends and drive KPIs.
- Built internal analytics tools (Streamlit/Flask) with interactive Plotly visualizations, automating workflows and improving stakeholder communication.
- Enhanced Name Screening and Mantas Alert Prioritization models by writing parameter optimization scripts, improving screening precision and alert accuracy.
- Automated the annual model review process (feature selection, model benchmarking, training period optimization), saving ~1 month of manual effort per review cycle.
- Built an LLM-powered BAU query handler for stakeholder inquiries on model scores, saving ~15 hours/week in manual response time.

PROJECTS

Document FAQ Generator - Multi-Agent RAG System | OCBC Bank Pte. Ltd, Singapore | May 2025 - December 2025

- Cut manual FAQ creation effort by ~80% and revision cycles by ~40% by building a multi-agent system (Summarizer, Indexer, FAQ Generator, Verifier) that transforms 100+ page PDFs into citation-backed FAQs.
- Improved retrieval precision by ~15% and reduced hallucinated answers by ~30% using FAISS-based vector search with optimized chunking and re-ranking.
- Improved average answer quality by ~20% over a single-model baseline by evaluating across multiple LLMs (Llama, Qwen, Gemma) with self-verification loops; containerized with Docker.
- Designed a Streamlit UI with chunk exploration, citation viewing, confidence signals, and JSON/CSV export for non-technical stakeholders.

ML Model Drift Review Multi-Agent System | OCBC Bank Pte. Ltd, Singapore | September 2024 - August 2025

- Reduced model drift review time from 1 hour to 10 minutes per model (~9 hours/week saved) by architecting a CrewAI-based system automating drift analysis across 9 production ML models.
- Built custom agents for MLflow data extraction, data cleaning, and explanation generation using Llama 3.1, leveraging function calling throughout the pipeline.

Invoice Extraction Engine - Multi-Model VLM + OCR | OCBC Bank Pte. Ltd, Singapore | November 2024 - July 2025

- Reduced invoice processing time from ~5 minutes to under 30 seconds (10x faster) by building an end-to-end extraction system for 1,000+ invoices using VLMs integrated with OCR.
- Achieved ~92-95% data accuracy across key fields using confidence scoring and rule-based validation, reducing correction errors by ~40%.
- Built Streamlit dashboards for preview, correction, and export, improving usability for business users.

Transaction Classification & Upsell Intelligence Platform | OCBC Bank Pte. Ltd, Singapore | December 2025 - June 2026

- Generated ~SGD 30K/month in additional revenue by building an LLM-powered transaction classification pipeline that surfaces personalized upsell opportunities for Relationship Managers.
- Designed an end-to-end pipeline: customers upload bank transactions, LLMs classify spend into segments (dining, travel, retail, etc.), statistical insights are derived, and LLMs generate personalized RM recommendations.
- Built an RM-facing interface to review customer insights and AI recommendations, reducing manual analysis time and improving pitch relevance and upsell conversion.

EDUCATION

Master of Computing, Specialization: Artificial Intelligence (OCBC SCHOLAR) | National University of Singapore | 2023 | GPA: 4.1

- Modules: Neural Networks and Deep Learning, Uncertainty Modelling in AI, Text Mining, AI Planning and Decision Making, Knowledge Discovery and Data Mining, Mining Web Data for Business Insights, Information Visualization, Digital Entrepreneurship.

Bachelor of Engineering (Computer Science) | BMSCE | Bangalore, India | 2019 | GPA: 8.8

- Coursework: Data Structures, Database Management Systems, Operating Systems, Computer Networks, Analysis and Design of Algorithms, Neural Networks.

SKILLS

Programming Languages: Python, C
ML Frameworks & Tools: PySpark, PyTorch, TensorFlow, Scikit-learn, XGBoost, LightGBM, Hugging Face, DGL, Pandas, NumPy, NLTK, Git, LangChain, CrewAI, OpenAI, FastAPI, Docker, Google Cloud Platform, MLflow, Tableau, Microsoft Excel, Apache Spark, Data Visualization.
Soft Skills: Teamwork, Communication, Time Management, Problem-solving, Stakeholder Management.